

Physics, 2017 EC, entrance exam, version 1

Grade 12 · Physics · 2017 E.C.

Time allowed: 120 minutes

Questions: 42

Total marks: 42

Name _____ Grade _____ ID No. _____ Date _____

1. If the tail of vector B is placed at the tip of A and the tail of C is placed at the tip of B, the sum of the three vectors is drawn from the: (1)
- A. Tail of A to tip of B
B. Tail of B to tip of C
C. Tail of A to tip of C
D. Tip of C to tail of B
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2. Which one of the following is not correct about the difference between distance and displacement? (1)
- A. Distance is the total length of path travelled by an object in a given time
B. Displacement is the shortest path between the initial and final positions.
C. Distance depends on the path followed by the object it moves.
D. Displacement can be more than or equal to the magnitude of distance
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3. A 25.0 kg block is initially at rest on a horizontal surface. A horizontal force of 75.0 N is required to set the block in motion. After it is in motion, a horizontal force of 60.0 N is required to keep the block moving with constant speed. Find the coefficients of static and kinetic friction respectively ($g = 10 \text{ m/s}^2$). (1)
- A. 0.1 & 0.2
B. 0.3 & 0.2
C. 0.2 & 0.3
D. 0.2 & 0.1
-
4. Which one is not true about action and reaction forces? (1)
- A. They cancel out each other
B. They act on two different bodies
C. They are same type
D. They exist in pairs
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5. A net force F_1 acting on a mass m causes an acceleration 'a'. When the force is tripled, the acceleration increased by 6. What will be the new acceleration acting on the same mass? (1)
- A. 6 m/s^2
B. 18 m/s^2
C. 12 m/s^2
D. 9 m/s^2
-
6. The magnitude of vector A is half of the magnitude of vector B. When they act parallel to each other, they yield a resultant of 90 units. The magnitudes of A and B respectively are: (1)
- A. A=60, B=30
B. A=37, B=53
C. A=30, B=60
D. A=53, B=37
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7. Which one of the following is not necessarily true? (1)
- A. All equal vectors are equivalent vectors.
B. All negative vectors are anti-parallel vectors.
C. Collinear vectors can be parallel or anti parallel to each other.
D. All coplanar vectors are collinear vectors
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8. What is the work done by a force $F = 20 \text{ N} (0.6i + 0.8j)$ in moving an object through displacement $S = 4 \text{ m} (2i + j)$? (1)
- A. $45(2.6i+1.8j) \text{ J}$
B. 160 J
C. 400 J
D. $200(1.4i+3j) \text{ J}$

9. The length of the train is 50 m and its front is 200 m from a pole. The train accelerates at 4 m/s^2 . At what speed does its front and rear pass the pole if it starts from rest? (1)

A. 80 m/s and $20\sqrt{5} \text{ m/s}$ B. 10 m/s and $20\sqrt{5} \text{ m/s}$
C. 40 m/s and $20\sqrt{5} \text{ m/s}$ D. 20 m/s and $20\sqrt{3} \text{ m/s}$

10. A cosmic body of extremely intensive gravity from which nothing, not even light can escape is (1)

A. Gravitational wave B. Higgs boson
C. Black hole D. Photons

11. The system we can apply for security lock in banking system to keep information secret from hackers and cyber-attacks is (1)

A. Quantum computing B. Quantum entanglement
C. Quantum cryptography D. Quantum teleportation

12. From the given graph, the body accelerated by 3 m/s^2 in the first 10 seconds and moves with constant velocity from 10 to 50 seconds and then decelerated by 5 m/s^2 from 50 to t seconds. What will be the acceleration between 10 to 50 seconds? (1)

A. 0 m/s^2 B. 3 m/s^2
C. 5 m/s^2 D. 8 m/s^2

13. Based on question no. 15 above, what is the total distance of the graph? (1)

A. 1200 m B. 1440 m
C. 2500 m D. 1080 m

14. Which of the following cannot be possible for the graphs of uniformly accelerated motion? (1)

A. The area under S-t graph B. The slope under V-t graph
C. The area under V-t graph D. The area under a-t graph

15. A man is weighing a fish of mass m using a spring balance attached to a ceiling of an elevator. Which one of the following statements is wrong about the reading of the balance? (1)

A. When the elevator is stationary its reading is mg
B. When the elevator is descending with acceleration a, its reading is $m(a+g)$
C. When the elevator acceleration a is equal to g, weightlessness happens
D. When the elevator is ascending up with acceleration a, reading is $m(a+g)$

16. A rectangular block weighs 600 N and has sides 3m, 2m and 4m. What is the minimum pressure the box exerts on the surface? (1)

A. 50 Pa B. 75 Pa
C. 100 Pa D. 150 Pa

17. What is the total pressure on a diver in a sea at a depth of 30 m below the surface? Density of sea water = 1020 kg/m^3 , $P_{\text{atm}} = 100,000 \text{ Pa}$, $g=10 \text{ m/s}^2$. (1)

A. 507 kPa B. 304 kPa
C. 205 kPa D. 406 kPa

18. The smaller piston of a hydraulic press has area 100 cm^2 while the larger piston has area 400 cm^2 . What force is exerted on the smaller piston to support a load of 100 N on the larger piston? (1)

A. 25 N B. 40 N
C. 250 N D. 400 N

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19. All are among the relationship of physics with biology except: (1)
- A. The fluid dynamics of blood flow through vessels
 - B. The mechanics of human motion
 - C. Atomic and molecular structure of minerals
 - D. The mechanisms for speaking and hearing
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20. Which of the following is false? (1)
- A. Air force demands concepts of physics about fluid mechanics, optics, navigation and fluid flow.
 - B. Defense technologies like Radar and Infrared night vision use physics electromagnetic waves.
 - C. Infrared imaging systems create an electronic image based on temperature difference.
 - D. Night vision image is blue to enhance natural night vision in humans.
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21. Sharper knives are effective in cutting. This is due to: (1)
- A. Higher pressure over a wider area
 - B. Higher pressure for a less area
 - C. Less force over a wider area
 - D. More force over a wider area
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22. Two balls are thrown with the same speed but at complementary angles. If one angle is 30° and its maximum height is h , then the maximum height of the other is: ($\sin 30=0.5$, $\cos 30=\sqrt{3}/2$) (1)
- A. $h/3$
 - B. $3h$
 - C. h
 - D. $2h$
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23. What is the final angular velocity when initial angular velocity is 5 rad/s , angular acceleration 2 rad/s^2 , time 10 s ? (1)
- A. 10 rad/s
 - B. 25 rad/s
 - C. 16 rad/s
 - D. 15 rad/s
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24. The gravitational force between two objects is $8 \times 10^{-9} \text{ N}$ when they are 3 cm apart. For what separation would the force be $6 \times 10^{-9} \text{ N}$ (same masses)? (1)
- A. $5\sqrt{2} \text{ cm}$
 - B. 5 cm
 - C. $2\sqrt{3} \text{ cm}$
 - D. 4 cm
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25. Which of the following is true about Kepler's law of planetary motion? (1)
- A. The line from sun to planet sweeps out different areas in equal times.
 - B. The square of the period is proportional to its average distance from the sun.
 - C. The period is proportional to its average distance from the sun.
 - D. The path is an ellipse with the sun at one focus.
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26. Which of the following is false? (1)
- A. Centripetal acceleration is towards the center.
 - B. Centripetal acceleration is perpendicular to tangential velocity.
 - C. Angular acceleration for uniform circular motion is tangent to the circle.
 - D. Radial acceleration arises from change in direction of velocity.
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27. What kind of systems do mechanical engineers create using concepts like dynamics and forces? (1)
- A. Medical devices
 - B. Chemical reactors
 - C. Robotics devices
 - D. Electrical circuits

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- 28.** Which of the following statements is correct? (1)
- A. An elastic collision: both kinetic energy and linear momentum are conserved.
 - B. An elastic collision: only kinetic energy is conserved.
 - C. Inelastic collision: both kinetic energy and linear momentum are conserved.
 - D. Inelastic collision: only kinetic energy is conserved.
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- 29.** An engineer wants to put a speed limit at the entrance to a train 300 m away as it comes to rest at the platform. He gives exactly 1.5 minutes for the train to decelerate uniformly within 300 m. What must be the maximum speed at the entrance? (1)
- A. 40 km/hr
 - B. 24 km/hr
 - C. 60 km/hr
 - D. 120 km/hr
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- 30.** A ball of mass 2 kg rolling at 10 m/s collides with another ball of mass 3 kg at rest and locks together. Common velocity after collision? (1)
- A. 2 m/s
 - B. 3 m/s
 - C. 4 m/s
 - D. 5 m/s
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- 31.** Two cars A and B moving same direction. Relative velocity of B wrt A is 10 m/s. If velocity of B is 60 km/h with respect to ground, what is velocity of A? (1)
- A. 35 km/h
 - B. 25 km/h
 - C. 55 km/h
 - D. 24 km/h
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- 32.** In which case is work done in physics? (1)
- A. A man carrying a load moving along a straight road
 - B. A man pushing a concrete wall
 - C. A man lifting a weight up onto a shelf
 - D. A man standing holding his bag
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- 33.** Which one cannot be the resultant when 10 N and 30 N are added by vector addition? (1)
- A. 15 N
 - B. 37 N
 - C. 20 N
 - D. 40 N
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- 34.** A body accelerates from 10 m/s to 20 m/s having mass 40 kg. Magnitude of work done? (ignore friction) (1)
- A. 60 J
 - B. 6000 J
 - C. 1200 J
 - D. 8000 J
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- 35.** A girl and boy are given similar work. If girl completes in half the time than the boy, which is true? (1)
- A. Boy is more powerful
 - B. Girl is more powerful
 - C. Equally powerful
 - D. Cannot determine
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- 36.** Which is correct about the four basic forces in nature? (1)
- A. Strong nuclear force is short-range and holds protons/neutrons together.
 - B. Weak nuclear force keeps electrons in orbit.
 - C. Electromagnetic force is the weakest.
 - D. Gravitational force acts at nuclear scale.

37. The unit vector in the direction of $A = 3i+4j$ is (1)

- A. $(3/2)i+2j$ B. $i+j$
C. $(3/5)i+(4/5)j$ D. $2i-j$

38. A driver moving at 20 m/s hits brakes, decelerates at 5 m/s². How far to stop? (1)

- A. 120 m
B. 100 m
C. 20 m
D. 40 m

39. Projectile fired at 53° with initial speed 10 m/s. Maximum height? ($\sin 53=0.8$, $\cos 53=0.6$) (1)

- A. 3.2 m B. 6.4 m
C. 5.6 m D. 7.2 m

40. A body thrown vertically upward at maximum height: (1)

- A.** Velocity zero, acceleration constant
- B.** Velocity zero, acceleration zero
- C.** Velocity zero, acceleration maximum
- D.** Velocity zero, acceleration minimum

41. Two balls X and Y have masses $M_X = 3M_Y$, dropped from same height at same time. Which reaches ground first? (1)

- A. X first
B. Y first
C. Both same time
D. Undetermined

42. If dot product of two vectors is zero, what do we know? (1)

- A.** Both zero vectors
B. They are unit vectors
C. They are parallel
D. They are perpendicular

— END OF EXAMINATION —